

Advanced Programming (CS4043)

Date: Sept 24th 2024

Course Instructor

Ms. Hira Butt

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 2

Roll No

Section

Student Signature

Do not write below this line

- Your Exam will be marked as Zero, if involved in any kind of cheating.
- Problem understanding is part of the exam, so no query will be entertained during the exam.
- This is A CLOSED notes/book exam. Any sheet is NOT ALLOWED.

PART 1 [4+3+3+4= 14 Marks]: Give short and precise answers to the following questions.

Q1: First guess what will be the output of this code snippet? If there is something wrong, explain the reason and provide a suitable solution for it, but if the output is fine, there is nothing to worry about 😊

```
const person = {  
  name: 'Alice',  
  greet: () => {  
    console.log('Hello, ${this.name}!');  
  }  
};  
  
person.greet();
```

Output: _____

Q2: Do you find any problem in this code? Highlight, explain the issue and correct it afterwards.

```
const PostDataComponent = () => {  
  // State for holding form data and response  
  const [responseData, setResponseData] = useState(null);  
  useEffect(() => {  
    // Function to handle the POST request  
    const dataToPost = { name: 'John Doe', age: 30 };  
    const response = axios.post('https://jsonplaceholder.typicode.com/xyz',  
      dataToPost).then(response=>{  
        console.log(response.data);  
      });  
    setResponseData(response.data);  
  }, []);
```

3: How can you create a new HTML element using JavaScript? Write code snippet only to create is "<h1>Hi this is me</h1>"

Q4: Scenario:

- The component initializes the `userName` state with the value from `props.name`.
- When the user clicks the button, it updates the `userName` to "New User".
- However, if the parent component updates `props.name` externally (e.g., via a prop update from a server or user input), the state `userName` will still be John or New User (if you pressed button before 3 seconds).
- If `props.name` changes later, `userName` will remain as whatever it was last set to and won't update, even when the parent component updates it? How would you fix the component so the child component reflects the latest name from props? Here is Parent Component too for your reference:
- Any other major issue in this existing code? Highlight and correct that too

```
const UserGreeting = (props) => {
  const [userName, setUserName] = useState(props.name);

  useEffect(() => {
    console.log("Rendering")
  }, [])

  const updateName = () => {
    setUserName('New User');
    props.name = "Jane";
  };

  return (
    <div>
      <h1>Hello, {userName}!</h1>
      <button onClick={updateName}>Update Name</button>
    </div>
  );
};

export default UserGreeting;

const ParentComponent = () => {
  const [name, setName] = useState('John');

  // simulate a change in the name prop after 3 seconds
  setTimeout(() => {
    setName('Jane');
  }, 3000);

  return <UserGreeting name={name} />;
};
```

2 [8+8=16 Marks]: Coding Part

Complete the following program to achieve the given tasks. Use declarative programming. Execution results must stay the same.

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product.filter(p => p.price)

Here is your dataset:

```
const products = [
  { id: 1, name: 'Laptop', category: 'Electronics', price: 1200, inStock: true },
  { id: 2, name: 'Headphones', category: 'Electronics', price: 80, inStock: false },
  { id: 3, name: 'Smartphone', category: 'Electronics', price: 900, inStock: true },
  { id: 4, name: 'Monitor', category: 'Electronics', price: 300, inStock: true },
  { id: 5, name: 'Book', category: 'Literature', price: 15, inStock: false },
  { id: 6, name: 'Chair', category: 'Furniture', price: 100, inStock: true },
  { id: 7, name: 'Table', category: 'Furniture', price: 250, inStock: true },
  { id: 8, name: 'Notebook', category: 'Stationery', price: 5, inStock: true },
  { id: 9, name: 'Pen', category: 'Stationery', price: 2, inStock: true }
];
```

Tasks:

Task 1: Most Expensive Product

Find the most expensive product in the dataset.

Return an object containing the product's ¹name, ²category, and ³price.

Task 2: Out of Stock Electronics

Identify all products in the "Electronics" category that are out of stock.

Return an array of product names.

Expected Result:

mostExpensiveProduct: { name: 'Laptop', category: 'Electronics', price: 1200 },
outOfStockElectronics: ['Headphones']

Instructions:

Use declarative programming techniques, such as map, filter, reduce, and any other relevant methods to accomplish each task.

Avoid using imperative constructs like for loops.

Provide a well-structured and organized solution for each task.

Here is a React App. Understand its flow and write solutions accordingly

Product Manager

Add/Update Product

Product ID	Product Name	Product Price	Add/Update Product
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Current Products

Laptop: \$1200
Smartphone: \$800
Headphones: \$150

When you fill all these details, it should add or update accordingly. If already exists, it should update the product only based on id. But if it is new, it will add it in current products list